

BASICS OF FIM · WI001287

Statistical Learning

0 SETUP

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University of Bayreuth
TUM School of Management

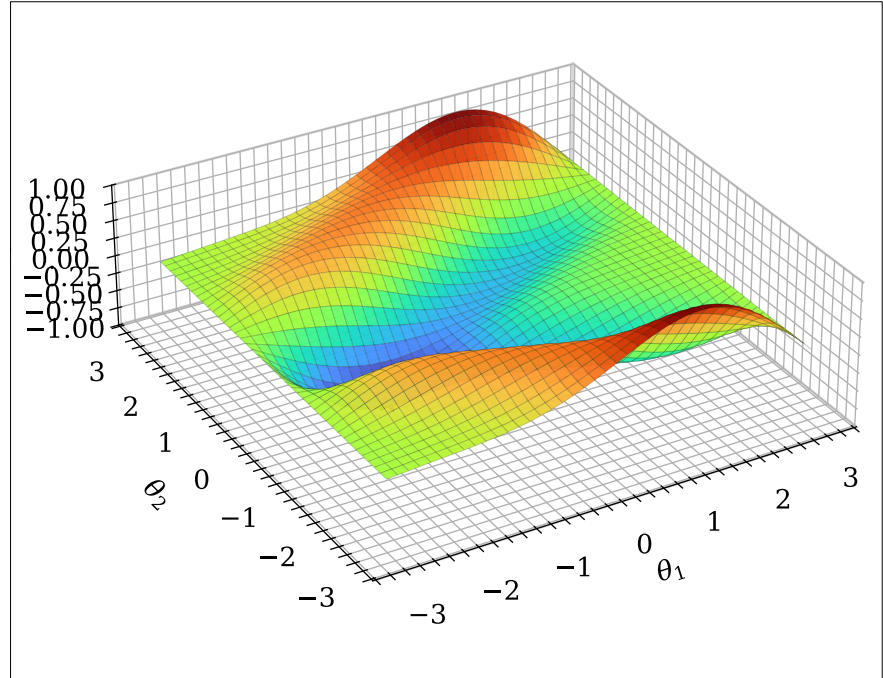


Figure 6.7: loss surface

Let's build our setup.



Python

Download
Installation



VS Code

Download
Installation
Extensions



Notebooks

Open folder in VS Code
Create an .ipynb file



Virtual Environment

Select Kernel
Run Notebook

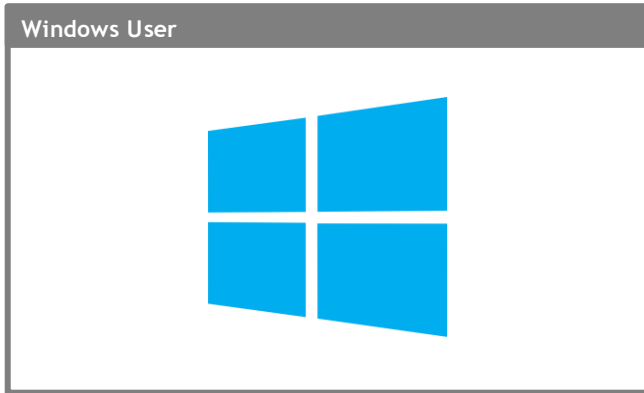
But first ...



... it's a decision...

But first ...

... are you a Windows or Mac user?



But first ...

... are you a Windows or Mac user?

Windows User



Team Blue

Mac User

Team Orange



But first ...

Windows User



Team Blue

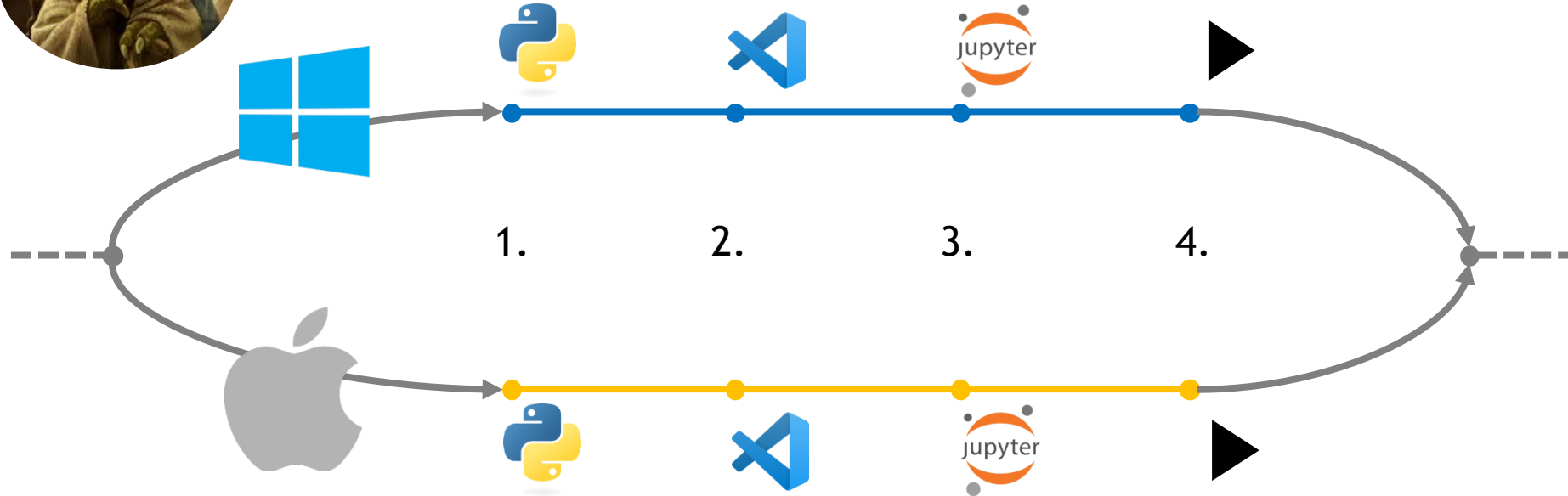
Mac User

Team Orange



... please help each other anyway!

Your path, you must follow.



Step 1: Creating our python environment

Step 1.1: Download miniconda installer

Visit the Anaconda website and download miniconda:

<https://www.anaconda.com/download/success>

Choose Your Download

Windows
Mac
Linux

Anaconda Distribution

Package and environment management with conda, Anaconda Navigator desktop app, and 600+ packages. Everything you need for data science.

GRAPHICAL INSTALLER

[↓ Windows 64-Bit Graphical Installer](#)

📖 Graphical install instructions

COMMAND LINE

📖 CLI install instructions

Miniconda

Lightweight installer that only includes conda, Python, and their dependencies. Install only what you need.

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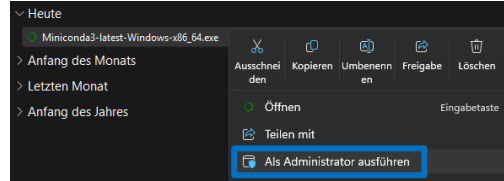
COMMAND LINE

📖 CLI install instructions

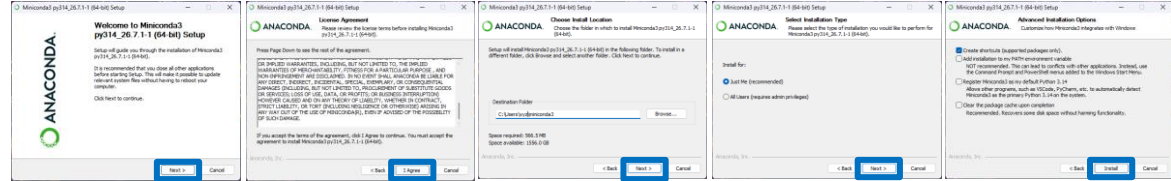
Step 1: Creating our Phyton environment

Step 1.2: Install miniconda (Windows)

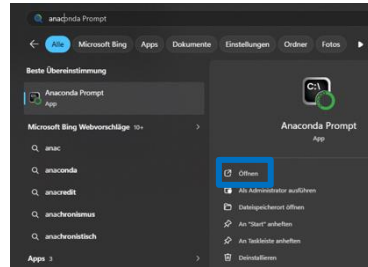
(i) Open the „exe“-file as administrator



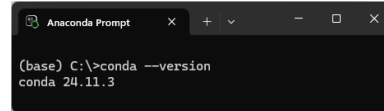
(ii) Click through the installation wizard



(iii) Search for “anaconda” in the windows-search and open it



(iv) Type “conda --version” to verify the installation. It should display something like “conda xx.xx.x”

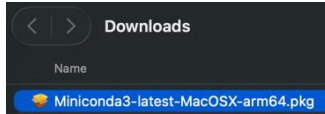


**Please note: While the interface is shown in German, the layout remains exactly the same.*

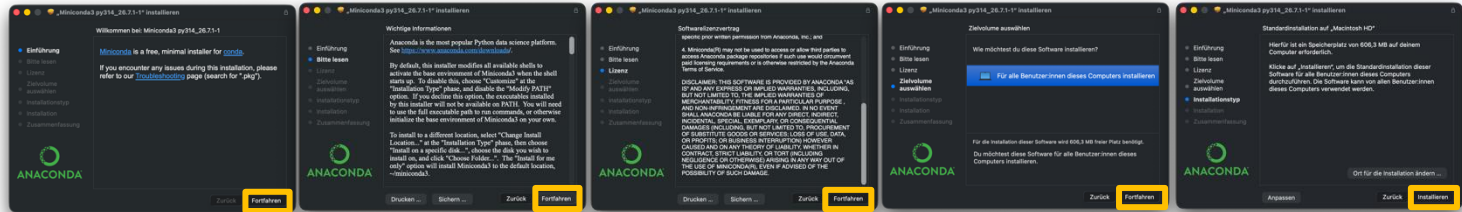
Step 1: Creating our Phyton environment

Step 1.2: Install miniconda (MacOS)

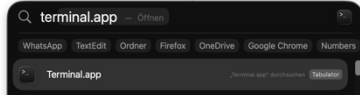
(i) Double-click the „.pkg“-file



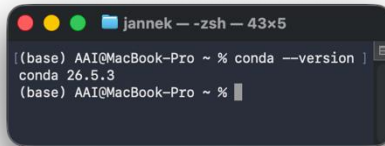
(ii) Click through the installation wizard



(iii) Open spotlight, search for “terminal” and open it



(iv) Type “conda --version” to verify the installation.
It should display something like “conda xx.xx.x”



**Please note: While the interface is shown in German, the layout remains exactly the same.*

Step 1: Creating our Python environment

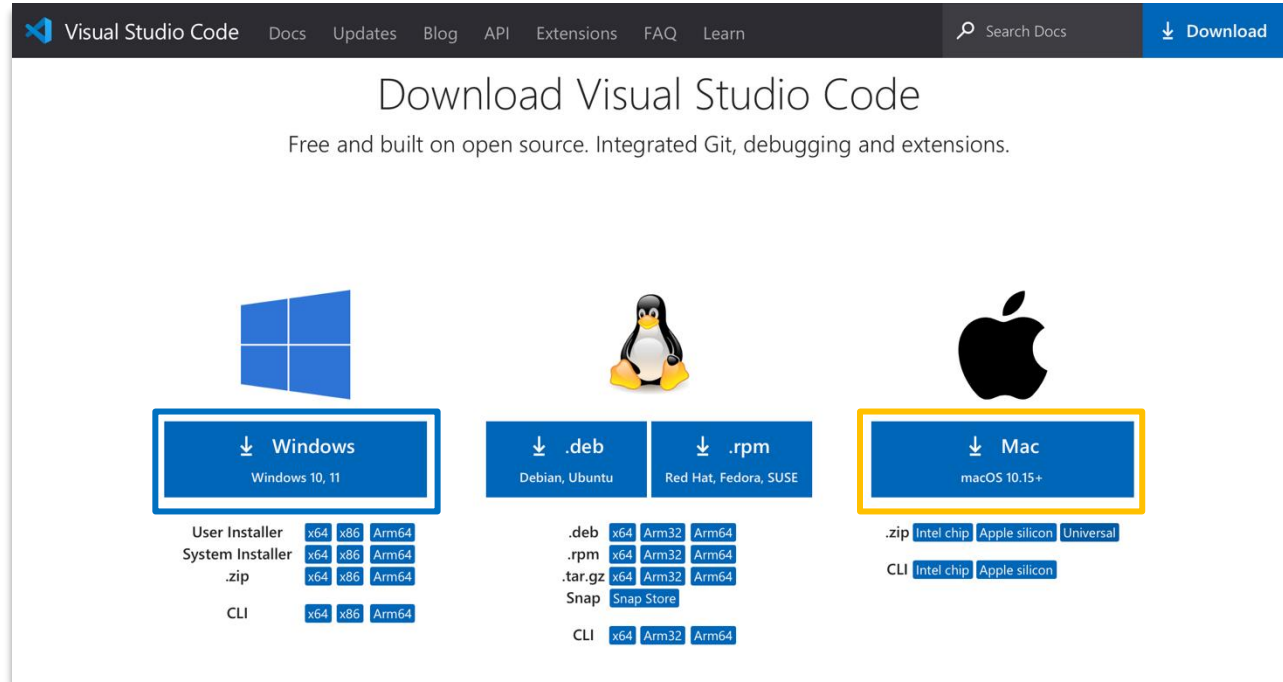
Step 1.3: Finally let's create our environment

This step is identical for [Windows](#) and [MacOS](#) users: Type in or copy the following lines of code and execute them line by line.

```
line 1  conda create -n AAI python=3.14 pip
line 2  conda activate AAI
line 3  pip install pandas matplotlib scikit-learn seaborn numpy scipy jupyter ipykernel
```

Step 2: Initiate the IDE

Step 2.1: Download and install VS Code



<https://code.visualstudio.com/Download>

Step 2: Initiate the IDE

Step 2.2: Open VS Code

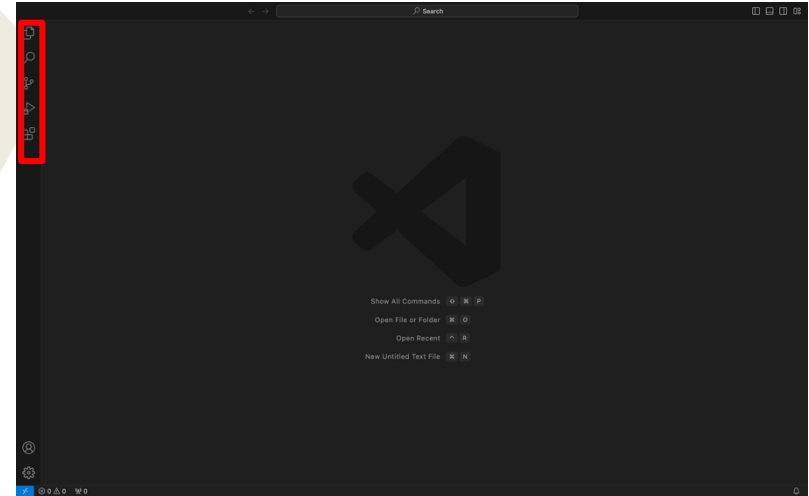
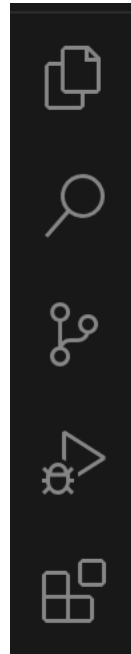
EXPLORER

SEARCH

SOURCE CONTROL

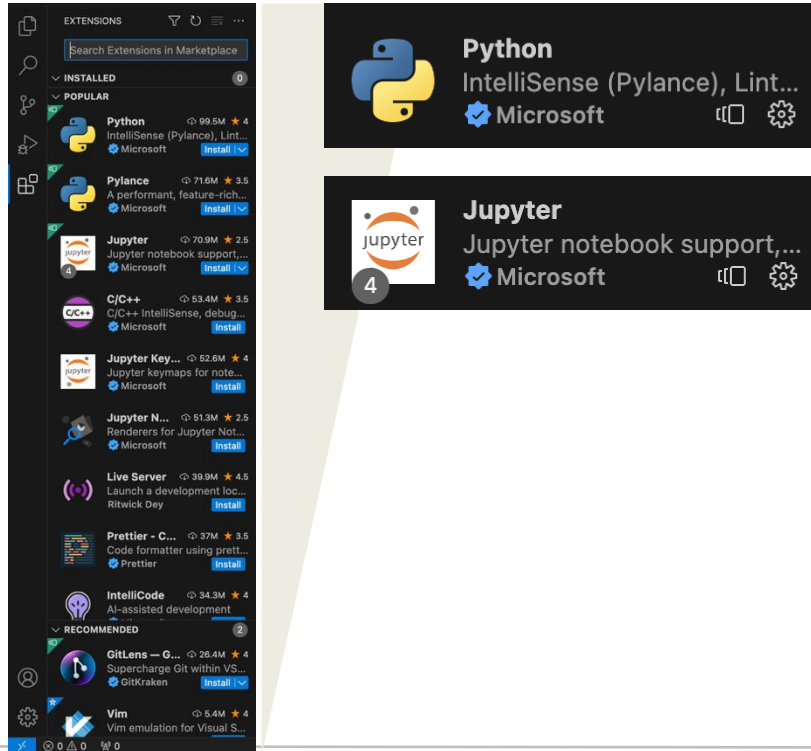
RUN AND DEBUG

EXTENSIONS



Step 2: Initiate the IDE

Step 2.3: Install VS Code Extensions



1. Python

2. Jupyter

Step 2: Initiate the IDE

Step 2.3: Install VS Code Extensions

Windows User

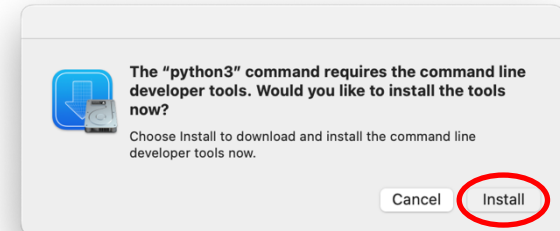
Install extensions

Install developer tools (shows up automatically)!

Mac User

Install extensions

Install developer tools (shows up automatically)!

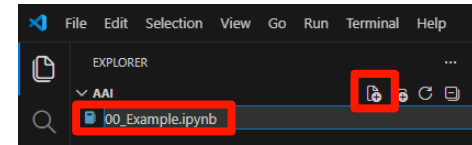
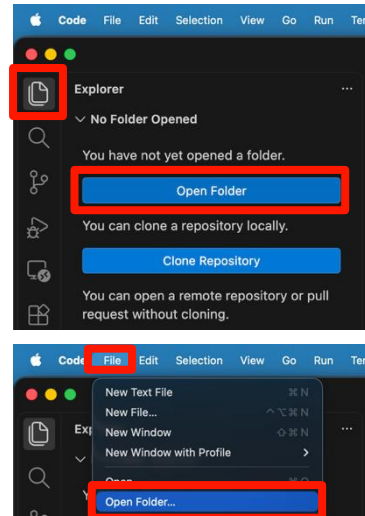
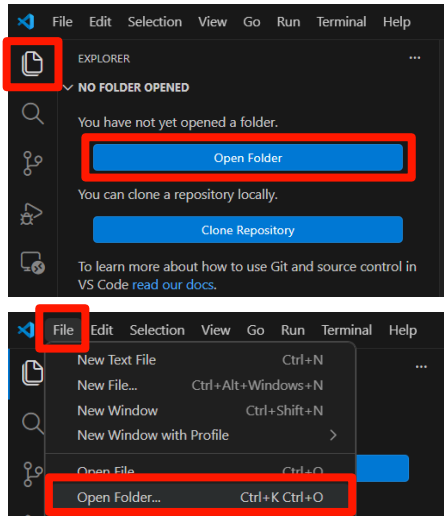


Step 3: Try out Python in VS Code

Step 3.1: Open a new folder and create your first notebook

(i) Go to the “explorer” tab and click “Open Folder” or click on “File” and “Open Folder...”.
The Layout is identical on **Windows** and **MacOS**.

(ii) Create a new file with the ending “.ipynb”
The Layout is identical on **Windows** and **MacOS**.

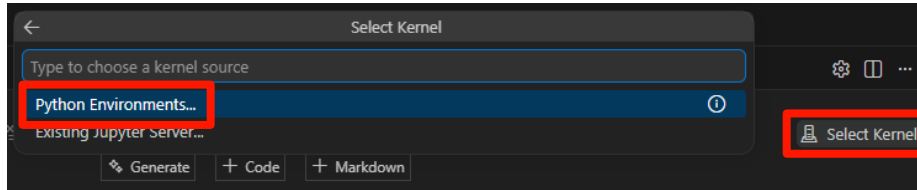


Please note: The next steps are identical on **Windows and **MacOS***

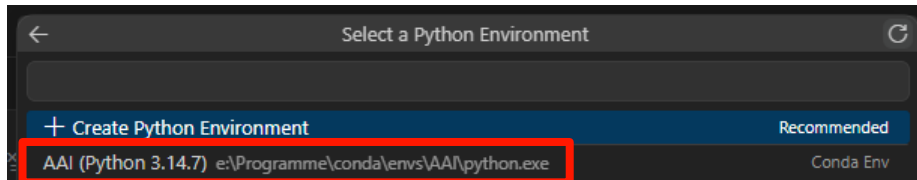
Step 3: Try out Python in VS Code

Step 3.2: Select our AAI environment

(i) In the created .ipynb file click on “Select Kernel” and “Python Environments...”



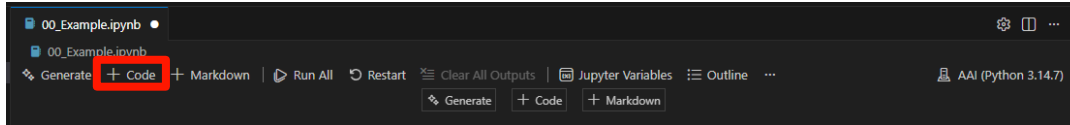
(ii) Now select our AAI environment



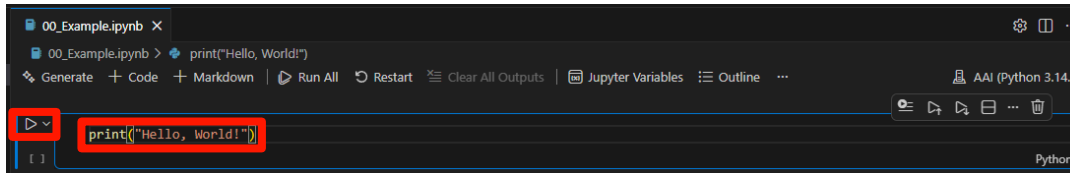
Step 3: Try out Python in VS Code

Step 3.3: Hello World example

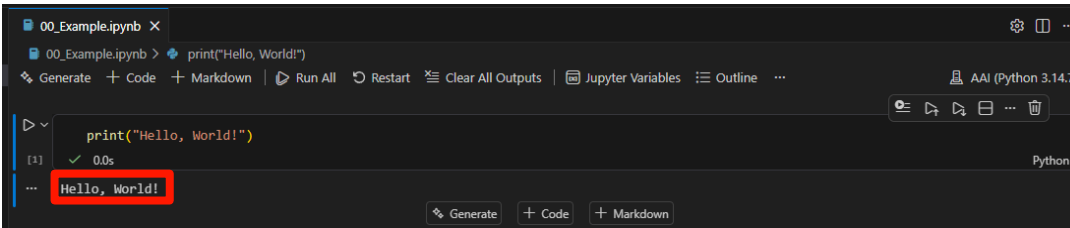
(i) Create a new “Code”-Cell



(ii) Type in the displayed code and click the run button



(iii) Look at the results



Some useful VS Code shortcuts (Windows)

<https://code.visualstudio.com/shortcuts/keyboard-shortcuts-windows.pdf>



General

Ctrl+Shift+P, F1	Show Command Palette
Ctrl+P	Quick Open, Go to File...
Ctrl+Shift+N	New window/instance
Ctrl+Shift+W	Close window/instance
Ctrl+,	User Settings
Ctrl+K Ctrl+S	Keyboard Shortcuts

Basic editing

Ctrl+X	Cut line (empty selection)
Ctrl+C	Copy line (empty selection)
Alt+↑/↓	Move line up/down
Shift+Alt+↑/↓	Copy line up/down
Ctrl+Shift+K	Delete line
Ctrl+Enter	Insert line below
Ctrl+Shift+Enter	Insert line above
Ctrl+Shift+V	Jump to matching bracket
Ctrl+]/[	Indent/outdent line
Home / End	Go to beginning/end of line
Ctrl+Home	Go to beginning of file
Ctrl+End	Go to end of file
Ctrl+↑/↓	Scroll line up/down
Alt+PgUp / PgDn	Scroll page up/down
Ctrl+Shift+I	Fold (collapses) region
Ctrl+K Ctrl+I	Fold (collapses) all subregions
Ctrl+K Ctrl+J	Unfold (uncollapses) all subregions
Ctrl+K Ctrl+O	Fold (collapses) all regions
Ctrl+K Ctrl+J	Unfold (uncollapses) all regions
Ctrl+K Ctrl+C	Add line comment
Ctrl+K Ctrl+U	Remove line comment
Ctrl+/	Toggle line comment
Shift+Alt+A	Toggle block comment
Alt+Z	Toggle word wrap

Navigation

Ctrl+T	Show All Symbols
Ctrl+G	Go to Line...
Ctrl+P	Go to File...
Ctrl+Shift+O	Go to Symbol...
Ctrl+Shift+M	Show Problems panel
F8	Go to next error or warning
Shift+F8	Go to previous error or warning
Ctrl+Shift+Tab	Navigate editor group history
Alt+←/→	Go back / forward

Ctrl+M Toggle Tab moves focus

Search and replace

Ctrl+F	Find
Ctrl+H	Replace
F3 / Shift+F3	Find next/previous
Alt+Enter	Select all occurrences of Find match
Ctrl+D	Add selection to next Find match
Ctrl+K Ctrl+D	Move last selection to next Find match
Alt+Ctrl+R / W	Toggle case-sensitive / regex / whole word

Multi-cursor and selection

Alt+Click	Insert cursor
Ctrl+Alt+↑/↓	Insert cursor above / below
Ctrl+U	Undo last cursor operation
Shift+Alt+I	Insert cursor at end of each line selected
Ctrl+L	Select current line
Ctrl+Shift+L	Select all occurrences of current selection
Ctrl+F2	Select all occurrences of current word
Ctrl+Shift+→	Expand selection
Shift+Alt+←	Shrink selection
Ctrl+Shift+Alt+←	Column (box) selection (drag mouse)
Ctrl+Shift+Alt+→	Column (box) selection
Ctrl+Shift+Alt+PgUp/PgDn	Column (box) selection page up/down

Rich languages editing

Ctrl+Space, Ctrl+I	Trigger suggestion
Ctrl+Shift+Space	Trigger parameter hints
Shift+Alt+F	Format document
Ctrl+K Ctrl+F	Format selection
F12	Go to Definition
Alt+F12	Peek Definition
Ctrl+K F12	Open Definition to the side
Ctrl+.	Quick Fix
Shift+F12	Show References
F2	Rename Symbol
Ctrl+K Ctrl+X	Trim trailing whitespace
Ctrl+K M	Change file language

Editor management

Ctrl+F4, Ctrl+W	Close editor
Ctrl+K F	Close folder
Ctrl+V	Split editor
Ctrl+1 / 2 / 3	Focus into 1st, 2nd or 3rd editor group
Ctrl+K Ctrl+←/→	Focus into previous/next editor group
Ctrl+Shift+PgUp / PgDn	Move editor left/right
Ctrl+K ←/→	Move active editor group

File management

Ctrl+N	New File
Ctrl+O	Open File...
Ctrl+S	Save
Ctrl+Shift+S	Save As...
Ctrl+K S	Save All
Ctrl+F4	Close
Ctrl+K Ctrl+W	Close All
Ctrl+Shift+T	Reopen closed editor
Ctrl+K Enter	Keep preview mode editor open
Ctrl+Tab	Open next
Ctrl+Shift+Tab	Open previous
Ctrl+K P	Copy path of active file
Ctrl+K R	Reveal active file in Explorer
Ctrl+K O	Show active file in new window/instance

Display

F11	Toggle full screen
Shift+Alt+O	Toggle editor layout (horizontal/vertical)
Ctrl+↵	Zoom in/out
Ctrl+B	Toggle Sidebar visibility
Ctrl+Shift+E	Show Explorer / Toggle focus
Ctrl+Shift+F	Show Search
Ctrl+Shift+G	Show Source Control
Ctrl+Shift+D	Show Debug
Ctrl+Shift+X	Show Extensions
Ctrl+Shift+H	Replace in files
Ctrl+Shift+J	Toggle Search details
Ctrl+Shift+U	Show Output panel
Ctrl+Shift+V	Open Markdown preview
Ctrl+V	Open Markdown preview to the side
Ctrl+K Z	Zen Mode (Esc Esc to exit)

Debug

F9	Toggle breakpoint
F5	Start/Continue
Shift+F5	Stop
F11 / Shift+F11	Step into/out
F10	Step over
Ctrl+K Ctrl+I	Show hover

Integrated terminal

Ctrl+`	Show integrated terminal
Ctrl+Shift+`	Create new terminal
Ctrl+C	Copy selection
Ctrl+V	Paste into active terminal
Ctrl+↑/↓	Scroll up/down
Shift+PgUp / PgDn	Scroll page up/down
Ctrl+Home / End	Scroll to top/bottom

Other operating systems' keyboard shortcuts and additional unassigned shortcuts available at aka.ms/vscodekeybindings

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⌘K ⇧[/ ⌘K ⇧]	Fold/unfold all subregions
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⇧⇧⇧ / ⇧⇧⇧	Expand / shrink selection
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⇧K ⇧⇧⇧ / ⇧K ⇧⇧⇧⇧	Move editor left/right
⇧K ⇧⇧ / ⇧K ⇧⇧⇧	Move active editor group

File management

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⇧O	Open File...
⇧S	Save
⇧⇧S	Save As...
⇧⇧S	Save All
⇧W	Close
⇧K ⇧W	Close All

⇧ET	Reopen closed editor
⇧K Enter	Keep preview mode editor open
⇧Tab / ⇧⇧Tab	Open next / previous
⇧K P	Copy path of active file
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